Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 3.0, **Version Date** 3/6/09

Linux Utilities

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1 Setting Date and Time

NOTE: If the time-zone is wrong, you must run `Reconfig` to select the correct one.

1. On the Service Desktop, select Utilities → Shutdown Applications.
2. Open a Shell and become <root> in the OC window.
3. Enter: **`su -`**

   Enter the super user (root) password.
4. Enter the following:

   ```
   root@hostname # setdate
   ```

   NOTE: Type **`q`** to quit anytime. Press <Enter> to proceed.

   To be accurate, this tool prompts you to enter the second. Watch your clock or PC carefully to enter the proper value, and press the <Enter> key at the right second to set the accurate time. Press the <Enter> key to proceed.

   ```
   Enter the current year (1980 -2030)<2007>
   Enter the current Month (1-12)<04>
   Enter the current Day (1-30)<14>
   Enter the current Hour (Military Time) (0-23)<18>
   Enter the current Minute (0-59)<13>
   Enter the current Second (0-59)<00>
   Updating the time on the OC and DARC, Please Wait...
   Current OC date: Fri Apr 14 15:18:00 CDT 2007
   Could not retrieve the DARC TimeZone.
   Please make sure the timeZones for the OC and DARC are the same.
   Setdate completed with NO ERRORS.
   [root@hostname]#
   ```
5. Close the shell by typing: **`exit`**
6. Type: **`st`** to restart application software.

2 Running storelog

Storelog is run when there is insufficient disk space on the OC disk to bring up the system to Applications Level. During Applications startup, the disk management process checks for adequate disk space; if there is insufficient space, **`storelog`** will come up automatically in a shell prompting the user. The threshold level for the OC that prompts the need for **`storelog`** is when the partitions approach 97-98% full.

Storelog is a tool that recovers system disk space by removing files not essential to the operation of the system. Storelog provides an option to save the files, (core, log, data) to DVD prior to removing them from the system disks.

Removing these “system log” files does not add image space, but should allow the applications to startup.

The storelog tool can be run standalone in a shell by simply typing **`storelog`**, or from the Service Desktop select [ERRORLOG], and choose [STORELOG].

3 Running sprsnap

The sprsnap script saves system information that is useful for debugging system problems. Information is saved to a directory in /usr/g/service/log, or to a DVD cartridge.

In addition to the available command switches for sprsnap, there is a command switch, **`-r`**, that gives an on-screen warning about not executing this program during patient data acquisition.

Usage: sprsnap -h, -s, -M, -i

- -h: describe usage of sprsnap
- -s: run in silent mode, no questions
- -a: run in fully automated mode (HAST)
- -M: save to DVD automatically, no prompt
- -i: save log files only
- -o: ignore file size threshold

FILES SPRSNAP SAVES

Core files from the OC

`/usr/g/service/log/core*`

`/usr/g/bin/core*`

`/usr/tmp/core*`

UNIX kernel core files from the OC

`/var/adm/crash/*`

`/usr/g/service/log` from the OC

`/var/adm/*SYSLOG*` files from the OC

`/var/adm/install*`

Install log files from the OC

`/usr/g/ctuser/logfiles/sdclog`

SDC log files from the OC

ScanRx info files from the OC

`/usr/g/service/log/exam*.protocol`

`/usr/g/service/log/exam*.scan.request`

`/usr/g/service/log/gesys_'uname -n'.log`

`/usr/g/queue`

Miscellaneous information such as disk space, process status and showprods in Queue directory.

Scan files

`/usr/g/data_manqagement/ex*/ex*_hdr`

SDC log from the OC

`usr/g/ctuser/logfiles/sdclog`

Miscellaneous files form OC

```
/usr/g/service/log/*.timers

/usr/g/vin/*.timers*

/usr/g/en_US/app_defaults/archive/SCSI.fol

/usr/g/en_US/app_defaults/devices/camera.dev

/usr/g/config/INFO

/usr/g/config/dataacq/cfg

/usr/g/config/dataacq.init

/usr/g/config/scan_file_mgr.cfg

/usr/g/config/scan_disk_io.dfg

/usr/g/config/scanfilemgr.init
```

4 Running quicksnap

Use quicksnap to save a snapshot of the system so engineering can analyze it and try to understand the cause of system performance problems, system crashes, etc. The data for quicksnap is saved to a directory in the following location: `/usr/g/service/log`. The directory is named as follows: `[hostname].[date-time string].inst.` For example, `msec7x2.06110294239.inst.`

To run quicksnap from the desktop:

1. From any desktop toolbox, click on the [Quick Snap] button. A popup warning is displayed.
2. Click on the [Ok]button on the warning screen. The screen remains on the monitor until the snapshot data collection is completed (usually under one minute).

To run quicksnap from a command line, open a UNIX shell and type one of the following commands:

- **`quicksnap -olc`** saves the snapshot data in the same manner as the desktop-based operation, but gives an on-screen indication of each step as it is performed.
- **`quicksnap -gui`** works the same as the desktop-based operation described previously.
- **`quicksnap -save`** gives the capability to select previously “snapped” data and save it to DVD (requires a blank, formatted DVD cartridge).

5 Initializing a MOD Cartridge

If you have a MOD upon which you want to put system files, this is different from image files. You prepare the MOD by making a file system on it. System State and DD File Analysis will detect this condition and prepare the MOD in the drive for you. To prep a system MOD under other circumstances, open a shell and type: **`mkfsMOD`** (formatting takes about 3-5 min).

NOTE: **`mkfsMOD`** will not work on DICOM image and options MODs.

6 Initializing a DVD

Initialize a blank DVD cartridge for the System State Save, quicksnap, or sprsnap programs to use.

1. Make sure the DVD cartridge is not write-protected before attempting to initialize it.
2. Open a UNIX shell and type: **`mkfsDVD`**

NOTE: If the DVD cartridge already has a file system on it, it warns the user before creating a new file system.

If the DVD cartridge has been initialized for storage of system options, the **`mkfsDVD`** script does not allow the user to reinitalize the cartridge.

7 Saving System State

To save the system configuration information, characterization, calibration, protocols, etc. to a System State DVD, perform the following tasks:

1. Bring the system up if it is not already up.
2. Insert the System State DVD (or MOD).
3. Click on the [SERVICE DESKTOP].
4. On systems with GRE (GOC3 or GOC4) consoles the System State selection is located under [UTILITIES].

   NOTE: Click on the [PM] icon.
5. Click on [SYSTEM STATE].
6. Click on [ALL]. This will highlight Cals, Characterization, Reconfig Info, etc.
7. Click on [SAVE].

   The save will take a few minutes. Review the output for errors or missing files; the scroll bar on the right works only when the tool isn't busy performing some task, it may take a little while.
8. Click on [DISMISS].

8 Saving and Restoring Scan Files

Saving scan files to DVD or MOD requires that the media first be formatted for a Linux (or UNIX) file system. Refer to Section 6 for formatting an DVD and to Section 5 for formatting an MOD. The Scan file save/restore function provides the ability to save/restore patient scan data (not images) to a DVD or MOD. This is used to allow engineering to analyze and troubleshoot unusual or intermittent data acquisition problems.

Saving scan files onto the MOD puts the scan files in the following directory path on the MOD: /`MOD/service_mod_data/SFfiles<suiteid>.exam.series.scan` file,. The scan files are saved as an iq. (referred to as a `.iq` file) in the `Sfiles` directory.

Restoring scan files function reads the MOD directory path: `/MOD/service_mod_data/Sfiles` and presents the files in a list/select GUI for restoring. Restoring the scan files takes a copy of `.iq` file on MOD and puts it into the scandata disk partition.

Both the Save and Restore Scan files functions reside under [RECONMGMT] on the top level ExamRx Desktop.

8.1 Saving Scan Files

Perform the following steps for Saving scan files:

1. Place a formatted DVD or MOD in the drive.
2. From the top level ExamRx Desktop, select [RECONMGMT].
3. Select [SAVE SCAN DATA].
4. Select the Exam, Series, Scan file(s) desired.
5. Select [SAVE SELECTED FILES].
6. When the save operation is complete, a pop-up reports "`xx scanfiles saved`" - select [OK].
7. Select [QUIT].

8.2 Restoring Scanfiles

Perform the following steps to restore scanfiles:

1. Place the DVD or MOD holding your scanfiles into the drive.
2. From the top level ExamRx Desktop, select [RECONMGMT].
3. Select [Restore Scan Data].

4. Select the Exam, Series, Scan file(s) desired.
5. Select [Restore Selected Scan Files].

9 Reserve/Release Scan Data

The reserve function allows you to prevent the overwriting of scan data files. The files can then be selected at a later time for future storage and/or reconstruction. Otherwise, eventually new scan data overwrites all of the existing scan data files. The release function unreserves any scan files previously reserved, freeing that scan file to be overwritten.

9.1 Reserving Scan Files

Perform the following steps for Reserving scan files:

1. From the top level ExamRx Desktop, select [RECONMGMT].
2. Select [RESERVE SCAN DATA].
3. A list of exam/series/scans currently unreserved is presented.
4. Click on the Exam/Series/Scan(s) desired to be reserved, then select [RESERVE SELECTED SCANFILES].
5. Operation is complete when a pop-up stating "`xx scanfile reserved`" comes up. Select [OK].
6. Select [QUIT].

9.2 Releasing Scan Files

Perform the following steps for Releasing scan files:

1. From the top level ExamRx Desktop, select [RECONMGMT].
2. Select [RELEASE SCAN DATA].
3. A list of exam/series/scans currently reserved is presented.
4. Click on the Exam/Series/Scan(s) desired to be released, then select [RELEASE SELECTED SCANFILES].
5. Operation is complete when a pop-up stating '`xx scanfile released`' comes up. Select [OK].
6. Select [QUIT].

10 Reinstall.db Script

NOTE: This utility does not destroy existing images.

Beginning with software version 05MW14.5, a new script for database management was created. The advantage of the reinstall.db script is that it reinstalls all images and recreates the database indexes, thus rebuilding the Terra database, instead of destroying the database and recreating it (as in resetImageDB_Linux).

Perform the following steps to use reinstall.db:

1. Shutdown the Applications *(Service Desktop>Utilities>Application Shutdown)*.
2. Open a Unix shell and type: **`reinstall.db`**. As the process runs, a screen displays the following message: `Please Wait ... Database is being reconstructed.`

NOTE:
- *Applications* must be shut down to use reinstall.db.
- The script reinstall rate is approximately 50 images/sec.
- The total time to reinstall all images is dependent on the database size.
- The CT host console is not available for other tasks during the database image reinstallation.
- The CT host console must be rebooted after the reinstall.db. script completes.

11 resetImageDB_Linux

NOTE: This utility destroys all images.

The resetImageDB_Linux command can be used on any of the current software. This command rebuilds the sdc_image_pool partition so all images are removed. sdc_image_pool partition is actually /dev/md0, which is a raid device created from striping two 73Gb disk drives together.

Perform the following steps to use resetImageDB_Linux:

1. Select [Shutdown], then select [Restart].

   NOTE: There have been some issues with the raid being lost if resetImageDB_Linux is run after shutting down applications and not rebooting. This is why a reboot is required prior to running resetImageDB_Linux.

2. Select [Cancel] when the attention: `You have 5 seconds to cancel Applications Startup` is displayed.

3. Open a Unix Shell and type **`su -<password>`**.

4. In the Unix Shell type: **`resetImageDB_Linux`**.

5. When prompted, type **`yes`** to continue with the database reset.

6. When resetImageDB is complete, type **`exit`** and **`st`** to start applications.

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